
Table 2.1 A Decathlon of Flourishing, from Age 60 to Age 80*

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1. Included in *Who's Who in America*
 2. Earning income in the Study's top quartile
 3. Low in psychological distress
 4. Success and enjoyment in work, love, and play since age 65 (Appendix D)
 5. Good subjective health at age 75 (that is, physically active at age 75)
 6. Good subjective and objective physical and mental health at age 80
 7. Mastery of the Eriksonian task of Generativity**
 8. Availability of social supports other than wife and kids between ages 60 and 75
 9. In a good marriage between ages 60 and 85
 10. Close to kids between ages 60 and 75
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* The Decathlon was conceived to assess the men's success from 65 to 80, but all but one of the events could be estimated for the 14 men who died between 58 and 64 years of age, and therefore they were included. Men who died before their 58th birthdays, however, were excluded.

** The men were coded for the highest Eriksonian psychosocial task they had mastered (see Chapter 5).

Table 2.2 What Variables Predict a High Decathlon Score at Age 60–80?

	Association with Total Decathlon Score
A. Variables Reflecting Constitutional Factors	
1. Mesomorphy (0 items*)	NS
2. Masculine body build (0 items)	NS
3. Vital affect (1 item)	Significant
4. Athletic prowess (2 items)	Very significant
5. Treadmill run (2 items)	Significant
6. Sociable/Extroverted (0 items)	NS
7. Mean ancestral longevity (1 item)	NS
8. Alcoholic relatives (2 items)	NS
9. Depressed relatives (0 items)	NS
10. Good childhood temperament (2 items)	Significant
B. Variables Reflecting Social Class	
1. Social class (upper-upper to blue collar) (0 items)	NS
2. Maternal education (6–20 years) (0 items)	NS
3. Paternal education (6–20 years) (0 items)	NS
C. Variables Reflecting Attachment	
1. Warm childhood (6 items)	Very significant
2. Overall college soundness (8 items)	Very significant
3. Empathic coping (defense)** style (age 20–35) (7 items)	Very significant
4. Warm adult relationships 30–45 (10 items)	Very significant

Very significant = $p < .001$; Significant = $p < .01$; NS = Not significant.

* “Items” refers to the number of Decathlon events predicted significantly by the antecedent variable.

** See Chapter 8.

Table 2.3 Childhood Predictors of Flourishing in Old Age

Decathlon of Flourishing	Environmental Childhood Strengths	Social Health Rating, Age 47	College Soundness, Age 21	Maturity of Defenses, Age 20–35
Total Decathlon score	Very Significant	Very Significant	Very Significant	Very Significant
<i>Who's Who in America</i>	NS	Significant	NS	Very Significant
Highest earned income	Significant	Very Significant	Significant	NS
Low lifetime distress	NS	Very Significant	Very Significant	Significant
Success and enjoyment in work, love, & play (age 65–80)	Very Significant	Very Significant	NS	NS
Subjective health (age 75)	NS	Very Significant	Significant	Significant
Successful aging (age 80)	NS	Very Significant	Significant	NS
Achieved Generativity	Very Significant	Very Significant	Very Significant	Very Significant
Social supports (other than wife and kids) (age 60–75)	Very Significant	Very Significant	Very Significant	Very Significant
Good marriage (age 60–85)	Significant	Very Significant	Significant	Significant
Close to kids (age 60–75)	Significant	Very Significant	NS	Significant

Very Significant = $p < .001$; Significant = $p < .01$; NS = Not Significant.

Table 3.1 The Historical Setting of the Grant Study Birth Cohort

	Personal Events	Historical/Public Events
1919–1922	Birth	End of World War I Women's suffrage in place
1923–1929	Childhood	Roaring Twenties Fathers getting richer Era of strict toilet training Fear that kissing children spreads disease
1930–1940	Adolescence	The Great Depression Fathers getting poorer Isolationism and pacifism
1941–1945	College	World War II
1946–1960	Military service Marriage Young families Career uncertainty	G.I. Bill Strong U.S. dollar Rapid growth Eisenhower years “Man in Grey Flannel Suit”
1961–1975	Consolidated careers Empty nests Leading and caring for adults	Vietnam Surgeon general's report on smoking Civil rights and Medicare U.S. dollar weaker End of Cold War
1990–2010	Retirement Poorer health Preserving culture	Clintons and Bushes Iraq War

Table 3.2 The Sequence of Contacts

1938–1945

8–10 psychiatric interviews
 Complete physical exam by Dr. Heath
 Interviews by Dr. Heath and Miss Gregory; she also made home visits
 Anthropological and physiological testing
 EEG, Rorschach, and handwriting analysis (on many of the men)
 Complete psychometric testing and some projective testing by Dr. Wells
 The 26 personality traits assigned by Dr. Woods (Appendix E)
 Brief childhood assessment (1–3) by Miss Gregory
 At age 21 the ABC College adjustment (soundness) ratings assigned

1946–1950

Debriefing of combat experience by Dr. Monks
 Annual questionnaires begun
 Interviews with wives by social anthropologist Dr. Lantis
 Thematic Apperception Test by Dr. McArthur
 Full staff conferences on each man at age 29; ABCDE ratings of personality soundness for future adjustment made

1950–1967

Mostly biennial questionnaires; little other contact

1967–1985

All men re-interviewed, mostly by Eva Milofsky or Dr. Vaillant (Appendix A)
 Beginning age 45, complete physical exams every five years until the present
 Objective health scored 1–5 (Well, minor illness, chronic illness, disabling illness, dead)
 Childhood environment assessed, blind to events after age 19
 Wives twice sent questionnaires
 Adjustment at work, love, and play assessed from age 30–47 on all men (Appendix D)
 Adaptive coping style (narcissistic, neurotic, empathic) assessed
 NEO (Costa and McCrae) administered by mail
 Lazare Personality Inventory administered by mail

1985–2002

All men re-interviewed (Appendix A)
 Adjustment at work, love, and play assessed from age 49–65 and from age 65–80 (Appendix D)
 Wives and children sent questionnaires
 Physical exams every five years and biennial questionnaires continued

Gallup Organization's Wellsprings of a Positive Life administered by mail
Aging at 80 computed (subjective and objective mental and physical health)

2002–2010

Physical exams every five years and biennial questionnaires continued
Joint marital interviews and “daily diaries”
Telephone Interview for Cognitive Status (TICS) at ages 80, 85, and 90
Re-interviewed at retirement, 1985–2005

2010

Decathlon scores compiled

Table 3.3 Responses to a Questionnaire Distributed at the 25th Reunion of the Harvard Class of 1944

	Grant Men N = 44	Classmates N = 590
Responded to reunion questionnaire	92% Significant	70%
Graduated from college with honors	61% Significant	26%
Went on to graduate school	76%	60%
Feels job is extremely satisfying	73% Significant	54%
Occupationally less successful than father	2% Significant	18%
Takes fewer than 2 days' sick leave a year	82% Significant	57%
Believes our involvement in Vietnam should decrease (winter 1968/69)	93%	80%
Attended public high school	57%	44%
Ever divorced	14%	12%
Often attends church	27%	38%
Drinks 4 shots (6 oz.) of liquor or more daily	7%	9%
10+ visits to a psychiatrist since college	21%	17%

Very significant = $p < .001$; Significant = $p < .01$; NS = Not significant.

Table 4.1 Important Associations of Heredity and Childhood Environment with Adjustment to Life

	Bleak Childhood Environment	Familial Mental Illness	Neuroticism / Extroversion Age 21
A. Mental Health			
Major depressive disorder	Significant	Very Significant	Significant
Psychiatric visits	NS	Significant	Very Significant
Lifetime mental illness	Significant	Very Significant	Very Significant
Adjustment, age 21	Very Significant	NS	NS
Maturity of defenses	Significant	Significant	NS
Adjustment, age 30–47	Very Significant	Significant	Significant
Decathlon	Very Significant	Significant	Very Significant
Adjustment, age 65–80	Very Significant	NS	Significant
B. Social Health			
Relationships, age 47	Very Significant	Very Significant	Significant
Social supports, age 50–70	Very Significant	NS	NS
Eriksonian maturation	Significant	Significant	Significant
Relationship with kids, age 50–70	Significant	NS	NS
C. Heredity More Important Than Childhood			
Alcoholism	NS	Very Significant	NS
Vascular risk variables*	NS	Significant	NS
Smoking	NS	Very Significant	NS

Very Significant = $p < .001$; Significant = $p < .01$; NS = Not Significant.

* This was a sum of five variables leading to vascular and heart disease (high diastolic blood pressure, Type II diabetes, obesity, heavy smoking, and alcohol abuse). The vascular risk variables are discussed in Chapter 7.

Table 6.1 Life Course of Study Marriages (1940 through 2010, or until the death of one partner)

MARITAL CATEGORY	N=237*	PERCENT
Still married; very happy intact first marriage**	51	21%
Divorced but happily remarried	23	10%
Still in a so-so intact first marriage	73	30%
Still in a poor intact first marriage	49	20%
Divorced; now single or unhappily remarried	39	16%
Never married	7	3%
Total	242	100%

*26 of the 268 original Study members were excluded from Table 6.1. Four of them died in the war, three never married, and 19 withdrew from the Study.

**One man who lived happily all his life with another man in a close interdependent relationship was classified as "Still married; very happy."

Table 6.2 Correlates of Good and Bad Marital Outcomes

Variable (n=159)★	Good marriages: n=50	Enduring bad marriages: n=48	Marriages that end in divorce: n=61
A. Good Marriages			
Warm father	38% (Significant)	21%	16%
Decathlon > 5	41% (Very Significant)	7%	5%
Mature defenses	42% (Very Significant)	16%	11%
Generative	74% (Very Significant)	46%	38%
B. Bad Marriages Or Divorce			
Lack of purpose and values	12%	11%	35% (Significant)
Poor social supports at 70	2%	47% (Very Significant)	50% (Very Significant)
Alcoholism in marriage	4%	46% (Significant)	57% (Very Significant)
Major depression	2%	27% (Significant)	20%
30+ days of tranquilizer use	12%	38% (Very Significant)	31% (Significant)
C. Surprisingly Not Significant			
Parents' marriage clearly stable	78%	54%	57%
Parents' marriage clearly unstable	14%	19%	24%
Deep religious involvement	20%	20%	24%
Bleak childhood	16%	38%	30%
Parents Catholic	14%	13%	10%

★ The total N in Table 6.2 is 159 rather than the 242 in Table 6.1 due to the exclusion of the 73 so-so marriages, the 7 never married, and 3 missing data.

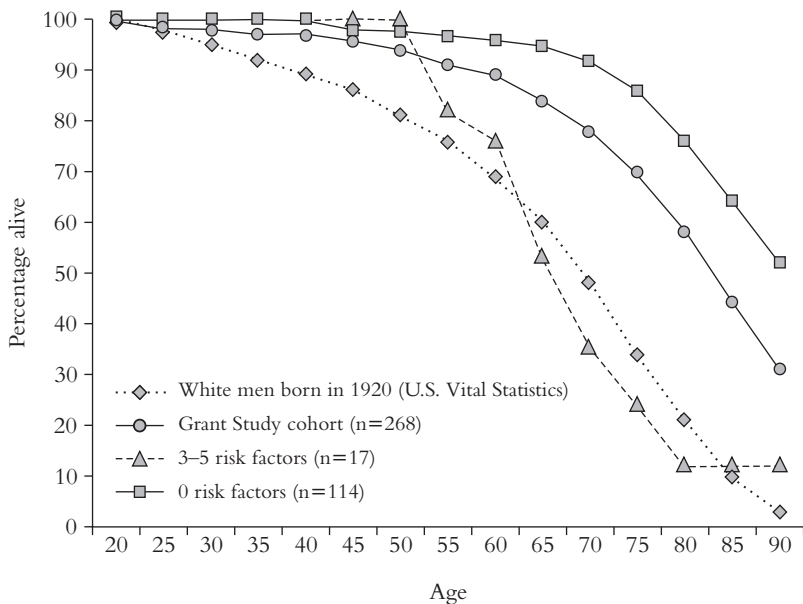


Figure 7.1 Percentage of men alive at each age.

Table 7.1 Instrumental Activities of Daily Living at Age 88 ± 1 (N=60★)

No cutbacks	12%
Can walk 2 miles without resting	41%
Still engages in hard physical activities: jogging, hiking, chopping wood, moving sofas	46%
Has not reduced a major activity	61%
Has not stopped a major activity	52%
Can still move light furniture or carry a suitcase through airports	74%
Needs no help going downtown	67%
Still drives and/or uses public transportation	75%
Can climb 2 flights of stairs without resting	79%
Needs no help with dressing or personal care	90%

★ Sixty people returned this questionnaire.

Table 7.2 Factors Associated with Irreversible Physical Health Decline Before Age 55

	Important Before Age 55 (n = 189)
A. Mental Health Variables	
High use of mood-altering drugs	Very Significant
Major depression	Significant
Bleak childhood	Very Significant
Poor adjustment 30-47	Significant
Poor relationships at 47	Very Significant
Immature defenses	Significant
B. Physical Health Variables	
Heavy smoking	NS
Alcohol abuse	Very Significant
Ancestral longevity	NS
Education	NS

Very Significant = $p < .001$; Significant = $p < .01$; NS = Not Significant.

Table 7.3 Factors Associated with Irreversible Physical Health Decline at Different Age Periods

	Important Before Age 55 (n = 189)	Important Between 55 and 80 (n = 177)	Important Between 80 and 90 (n = 155)
A. Mental Health Variables			
High use of mood-altering drugs	Very Significant	NS	NS
Major depression	Significant	NS	NS
Bleak childhood	Very Significant	NS	NS
Poor adjustment 30-47	Significant	NS	NS
Poor relationships at 47	Very Significant	NS	NS
Immature defenses	Significant	NS	NS
B. Physical Health Variables			
Vascular Risk Factors:			
a. Smoking	Very Significant	Very Significant	Very Significant
b. Alcohol abuse	NS	Significant	NS
c. High diastolic blood pressure at 50	Very Significant	Very Significant	NS
d. Obesity	Very Significant	Very Significant	Significant
e. Type II diabetes	Very Significant	Very Significant	NS
Ancestral Longevity	NS	NS	NS
Education	NS	NS	NS
Cognitive Competence at 80	NS	Significant	NS
Diagnosis of Cancer	NS	Very Significant	Very Significant

Very Significant = $p < .001$; Significant = $p < .01$; NS = Not Significant.

Table 7.4 Factors Associated with Longevity (Mean Age at Death 81.5; N = 237)

	N	Mean Age	Statistical Significance
A. Significantly Shortened Longevity			
Alcohol dependence	18	67.6	Very Significant
3+ vascular risk factors	17	68.3	Very Significant
Heavy smoker	40	73.0	Very Significant
High combat	16	75.3	Significant
Neither well integrated nor self-driving	83	78.8	Significant
Only a college degree	81	79.0	Very Significant
Bottom sixth in ancestral longevity	36	77.1	Significant
B. Significantly Lengthened Longevity			
No vascular risk factors	113	86.4	Very Significant
Never smoked	77	84.7	Very Significant
Good adjustment age 30–47	58	85.1	Very Significant
Low diastolic blood pressure at 50	66	85.9	Very Significant
Top sixth in ancestral longevity	44	84.3	Significant
Both well integrated and self-driving	23	89.3	Significant
Graduate school	153	83.3	Very Significant
C. Did Not Significantly Affect Longevity			
Warmest childhoods	57	83.0	NS
Bleakest childhoods	61	80.0	NS
Cholesterol above 253 mg/dl	57	81.1	NS
Cholesterol below 206 mg/dl	57	82.9	NS
Low exercise (age 20–45)	79	80.4	NS
High exercise (age 20–45)	25	85.0	NS
Parental social class: Upper	80	82.4	NS
Parental social class: Blue collar	14	81.6	NS
No psychosomatic problems	50	82.0	NS
2+ psychosomatic problems	48	83.7	NS
No obesity	212	82.2	NS
Obese (BMI 28+)	23	79.3	NS
Mature defenses	39	84.0	NS
Immature defenses	34	78.4	NS
Highest social supports	54	84.1	NS
Lowest social supports	59	80.1	NS

Very Significant = $p < .001$; Significant = $p < .01$; NS = Not Significant.

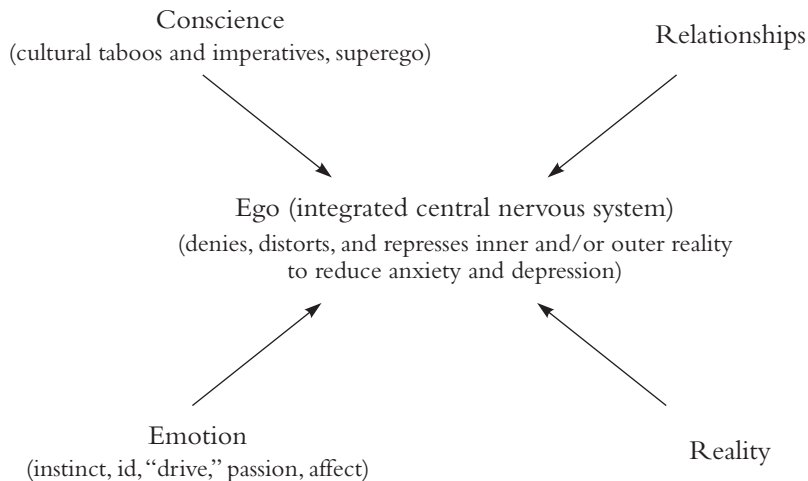


Figure 8.1 The four sources of intrapsychic conflict.

Table 8.1 Maturation of Defenses Over the Lifespan

	Adaptive Category Assigned to Vignettes★		
	“Immature”	“Intermediate/Neurotic”	“Mature”
Adolescence	25%	61%	14%
19–35	12%	58%	30%
35–50	9%	55%	36%
50–75	6%	32%	62%

★The vignettes are the ministories excerpted from the men's histories as examples of adaptive coping behavior, and then identified and rated.

Table 8.2 Correlations Between Maturity of Defenses and Measures of Successful Adult Outcome

Variables	Terman women n = 37	College men n = 186	Inner City men n = 307
Life satisfaction, age 60–65	.44 Significant	.35 Very Significant	n.a.
Psychosocial maturity (Erikson stage)	.48 Very Significant	.44 Very Significant	.66 Very Significant
DSM-IV Axis V	.64 Very Significant	.57 Very Significant	.77 Very Significant
Job success, age 47	.53 Very Significant	.34 Very Significant	.45 Very Significant
Marital stability, age 47	.31 Significant	.37 Very Significant	.33 Very Significant
Job enjoyment, age 47	.51 Very Significant	.42 Very Significant	.39 Very Significant
% of life employed	.37 Significant	n.a.	.39 Very Significant

Very Significant = $p < .001$; Significant = $p < .01$; NS = Not Significant.

n.a. = data not available.

Table 8.3 Late Life Consequences of Mature Defenses at Age 20–47 for the College Cohort

	n = 154★
I. Objective Evidence	Adaptiveness of Defenses
Income (50–55)	Very Significant
Psychosocial adjustment (50–65)	Very Significant
Social supports	Very Significant
Decathlon (60–80)	Very Significant
Mental health (64–80)	Very Significant
Good marital outcome (50–85)	Very Significant
II. Subjective Evidence	
Joy in living (75)	Very Significant

Very Significant = $p < .001$; Significant = $p < .01$; NS = Not Significant.

★ Sample size reflects the fact that we did not have information on all the men on all the variables.

Table 8.4 Use of Defenses by the Most Depressed and Least Distressed Men in the College Sample

Defenses	Percent Using Each Defense as a Major Coping Style		Significance
	Most Depressed (n = 17)	Least Distressed (n = 59)	
Suppression	18%	63%	Very Significant
Altruism	0%	19%	NS
Reaction formation	28%	2%	Significant
Dissociation	59%	25%	Significant
Projection	24%	3%	Very Significant
Most mature	0%	31%	Very Significant
Least mature	53%	9%	Very Significant

Very Significant = $p < .001$; Significant = $p < .01$; NS = Not Significant.

Table 8.5 Correlations Between Maturity of Defenses and Biopsychosocial Antecedents

Background variable	Terman Women (n = 37)	College Men (n = 186)	Inner City Men (n = 277)*
Parental social class	NS	NS	NS
IQ	NS	NS	NS
Years of education	NS	—	Significant
Warm relations with father	NS	Significant	NS
Warm childhood environment	Significant	Significant	NS
Warm relations with mother	NS	Significant	NS

Very Significant = $p < .001$; Significant = $p < .01$; NS = Not Significant.

*Thirty men with IQs less than 80 were excluded.

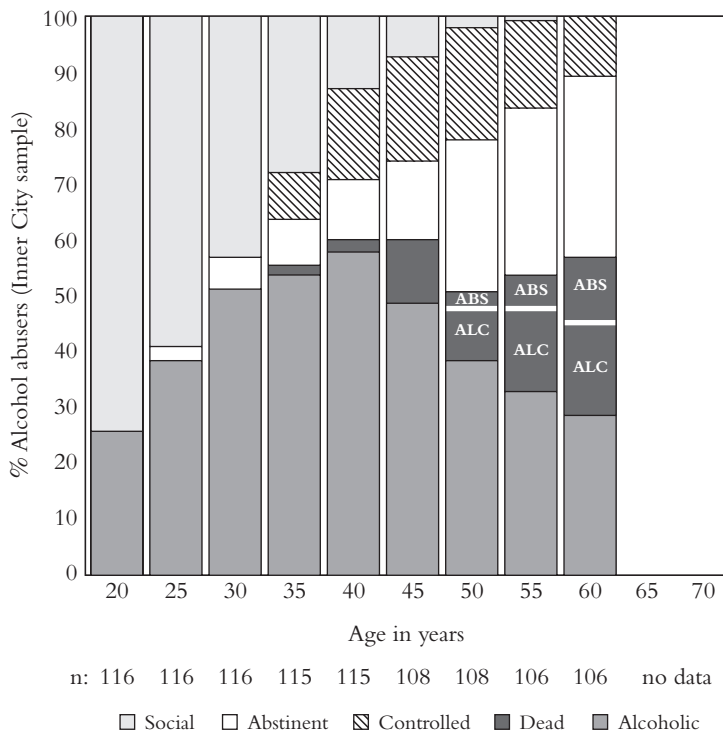


Figure 9.1 Final alcohol outcome status.

Table 9.1 Correlation of Alcoholism and Dependent Traits with Premorbid Variables

	Dependent Traits N = 95	Alcohol Abuse N = 185
Family history of alcoholism	NS	Significant
Poor childhood environment	Very significant	NS
Poor father-child relationship	Significant	NS
Poor high school social adjustment	Significant	NS
Poor college "psychological soundness"	Significant	NS

Very significant = $p < .001$; Significant = $p < .01$; NS = Not Significant.

Table 9.2 Outcome Status for Study Alcoholics at Age 70 or at Time of Death

	College Cohort				Inner City Cohort			
	In Community at Age 70		Dead at Age 70		In Community at Age 70		Dead at Age 70	
	N	%	N	%	N	%	N	%
Stable abstinence (3+ years)	4	14%	5	26%	34	64%	17	36%
Return-to-controlled-drinking (3+ years)	5	17%	2	11%	3	6%	8	17%
Chronic alcohol abuse	20	69%	12	63%	16	30%	22	47%
Total	29	100%	19	100%	53	100%	47	100%

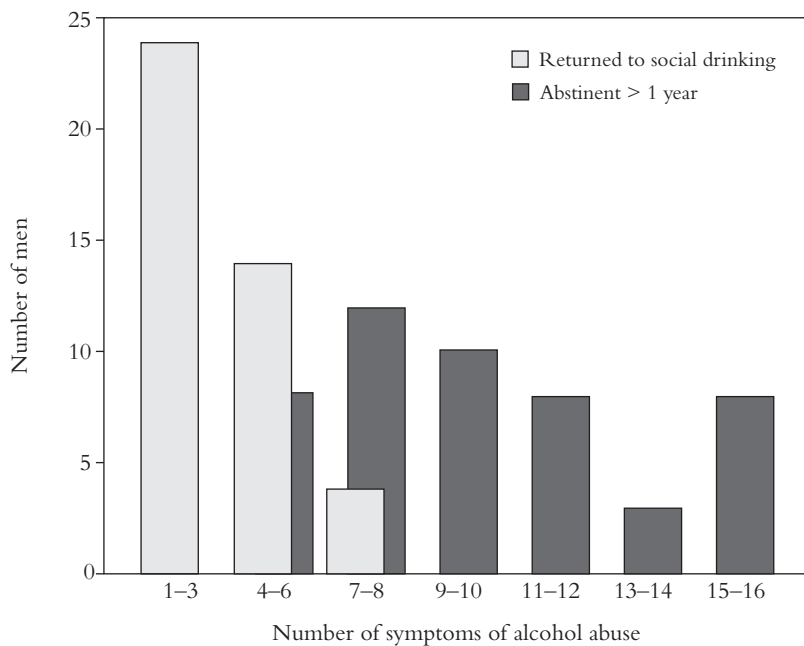


Figure 9.2 Relation between number of alcohol-related problems and likelihood of a return to social drinking.

Table 9.3 Factors Associated with Abstinence for a Year or More for 49 Untreated and 29 Clinic-Treated Inner City Men

	Untreated Abstinent Alcoholics n = 49	Treated Abstinent Alcoholics n = 29
Compulsory supervision	49%	34%
Substitute dependence	53%	55%
New relationships	32%	31%
Inspirational group membership (usually AA)	49%	62%

Table 9.4 Question: How Does AA Work? Answer: AA Works Fine!

	COLLEGE COHORT		INNER CITY COHORT	
	Abstinent n = 9	Chronic n = 32	Abstinent n = 57	Chronic n = 44
Average years of abstinence	15 years	1 year	16 years	1 year
Years of active alcoholism	20 years	23 years	18 years	22 years
Problem drinking score	9	6 (VS)	10	8
AA meetings	137	2 (VS)	143	8 (VS)

Very significant = $p < .001$; Significant = $p < .01$; NS = Not Significant.

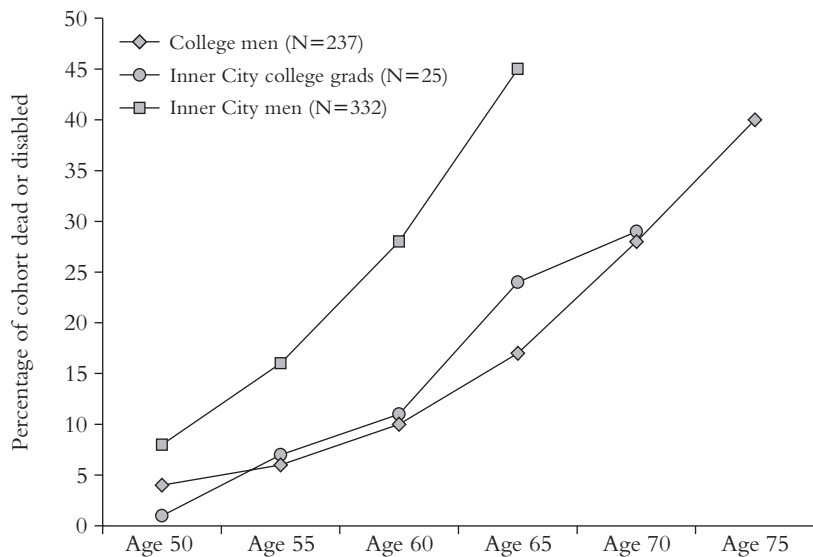


Figure 10.1 Death after fifty for College men, Inner City men, and Inner City college graduates.

Table 10.1 Religious Involvement and Adult Adjustment

	Religious Involvement, age 45–75 N = 224	Adult Adjustment, age 50–65 N = 224
Mental Health		
Childhood social class	Not significant	Not significant
Warm childhood	Not significant	Significant
Psychological soundness (age 21)	Not significant	Significant
Warm relationships (age 47)	Not significant	Very significant
Warm marriage (age 50–70)	Not significant	Very significant
Close to kids (age 50–70)	Not significant	Very significant
Social supports (age 55–75)	Not significant	Very significant
Adult adjustment (age 50–65)	Not significant	—
Neuroticism (age 60)	Not significant	Very significant
Smoking (age 20–60)	Not significant	Very significant
Alcohol abuse (age 20–65)	Not significant	Very significant
Physical Health		
Objective health (age 45)	Not significant	Very significant
Objective health (age 60)	Not significant	Very significant
Objective health (age 70)	Not significant	Very significant
Years disabled or dead <80	Not significant	Very significant

Very significant = $p < .001$; Significant = $p < .01$

Table 10.2 Spirituality and Warm Relationships

	Religious Involvement Age 45–75 n = 91–98	Adult Adjustment Age 50–65 n = 91–98
Faith-based beliefs:		
I believe that each person has a soul	Very significant	Not significant
I believe that people are born to do good	Significant	Not significant
I believe in a universal power, a god	Very significant	Not significant
In the last 24 hours, I have spent 30 minutes in prayer, meditation, or contemplation	Very significant	Not significant
Pragmatic beliefs:		
I believe I am the most important person in someone else's life	Not significant	Very significant
I have cared deeply about someone for ten or more years	Not significant	Significant
I always feel the presence of love in my life	Not significant	Very significant

Very significant = $p < .001$; Significant = $p < .01$; NS = Not significant.

Table 10.3 Mean Age at Death of Maternal Grandfather and of Other Primary Ancestors in the Four Affective Disorder Categories

Affective Disorder Category	Maternal Grandfathers, Mean Age of Death	Other 5 Ancestors, Mean Age of Death
Major depressive disorder (n = 23)	60 Very significant	71 NS
Personality disorder/alcoholism (n = 35)	66 Significant	72 NS
Intermediate (n = 114)	69 NS	74 NS
“Undistressed” (n = 58)	75 Very significant	73 NS
Total sample (N = 230)	69	73

Very significant = $p < .001$; Significant = $p < .01$; NS = Not significant.

Table 11.1 Statistical Strength of Association of Alternative Predictors of Flourishing*

Decathlon of Flourishing	Maturity of Defenses, Age 20–47	Warm Childhood Environment	Extraversion Minus Neuroticism, Age 21	Extraversion Minus Neuroticism, Age 68
Decathlon	Very Significant	Very Significant	Very Significant	Very Significant
<i>Who's Who in America</i>	Very Significant	NS	Very Significant	NS
Highest earned income	NS	NS	Very Significant	Significant
Low lifetime distress	Very Significant	NS	Significant	Very Significant
Success and enjoyment at work, love and play (age 65–80)	Very Significant	Very Significant	Significant	Very Significant
Generativity achieved	Significant	Significant	Significant	Very Significant
Subjective health (age 75)	Very Significant	NS	NS	Significant
Successful aging (age 80)	Significant	NS	NS	Very Significant
Social supports (other than wife and kids) (age 60–75)	Very Significant	NS	NS	Significant
Good marriage (age 60–85)	Significant	NS	NS	Very Significant
Close to kids (age 60–75)	NS	Significant	NS	NS

* 168 men were included in this table. Only the classes of 1942–1944 were rated for defenses, and some men have been lost to death.

Very Significant = $p < .001$; Significant = $p < .01$; NS = Not Significant.

APPENDIXES

APPENDIX A: THE INTERVIEW SCHEDULES

AGE 45–55

AGE 65–80

APPENDIX B: THE INNER CITY MEN AND THE Terman WOMEN

APPENDIX C: ASSESSMENT OF CHILDHOOD SCALES

APPENDIX D: ADULT ADJUSTMENT SCALES

AGE 30–50

AGE 50–65

AGE 65–80

APPENDIX E: DOMINANT COLLEGE PERSONALITY TRAITS

APPENDIX F: STUDY BIBLIOGRAPHY

APPENDIX A

THE INTERVIEW SCHEDULES

The Grant Study Interviews. Below are two examples of the semi-structured schedules of questions I used to guide my interviews with the men. Within reason, listed questions were always asked in the same order. The interviewer took longhand notes during the interview. Whenever a question elicited a problem area in a man's life, the interviewer probed for his particular coping style.

The Interview Schedule, age 45–55

I. Work

- a. What do you do? Any recent changes in responsibility?
- b. Ten years from now, where are you heading?
- c. What do you like and what do you dislike about your work?
- d. What for you is most difficult?
- e. What job would you have preferred?
- f. What are good and bad aspects about relations with your boss? With subordinates?
- g. How do you handle some of the problems that arise with these people?
- h. Looking back, how did you get into your present work?
 - i. Were there people with whom you identified?
 - j. What work do you do outside of your job—degree of responsibility?
- k. What plans for retirement?
- l. Ever unemployed for more than a month? Why?

- m. What will you do the first week of retirement? Anticipated feelings?

II. Family

- a. News of parents and siblings.
- b. Describe each child, their problems and sources of concern to you.
- c. How do you handle adolescence differently from your parents?
- d. Any deaths: first response, second response, means of finally handling feelings.
- e. This is the hardest question that I shall ask: Can you describe your wife?
- f. Since nobody is perfect, what causes you concern about her?
- g. Style of resolution of disagreements.
- h. Has divorce ever been considered? Explain.
- i. Quality of contact with parents and degree of pleasure derived.
- j. Which of your parents wore the pants when you were younger?

III. Medical

- a. How is your health overall?
- b. How many days sick leave do you take a year?
- c. When you get a cold, what do you do?
- d. Specific medical conditions and means of coping with the disability.
- e. Views and misconceptions about these conditions.
- f. Injuries and hospitalizations since college.

- g. Patterns of use of smoking and recollections about stopping.
- h. Pattern of use of medicines and of alcohol.
- i. Do you ever miss work due to emotional strain, fatigue, or emotional illness?
- j. Effect of work on health and vice versa.
- k. How easily do you get tired?
- l. Effect of health on the rest of your life.

IV. Psychological

- a. Biggest worries last year.
- b. Dominant mood over past six months.
- c. Some people have trouble going for help and advice: What do you do?
- d. Can you talk about your oldest friends? What made them friends?
- e. Who are the people (non-family) you would feel free to call on for help?
- f. What social clubs do you belong to, and what is your pattern of entertaining?
- g. How often do you get together with friends?
- h. What do people criticize you for or find irritating about you?
- i. What do they admire or find endearing?
- j. What are your own satisfactions and dissatisfactions with yourself?
- k. Ever seen a psychiatrist? Who? When? How long? What do you remember? What did you learn?
- l. Persistent daydreams or concerns that you think about but don't tell others?
- m. Effect of emotional stress?

- n. Philosophy over rough spots?
- o. Hobbies and use of leisure time? Athletics?
- p. Vacations? How spent and with whom?
- q. Any questions raised by my review of case record?
- r. What questions do you have about the Study?

The Interview Schedule, age 65–80

I. Work

1. Why did you retire? When?
2. How was your job over the last two years? What did you miss most? Was there a retirement ceremony?
3. What did you do the first six weeks of retirement?
4. What takes the place of your job?
5. What is your most important activity now?
6. What is the best part of retirement?
7. The worst part?
8. The most difficult part?
9. How is your retirement financed?
10. Any job change if you were to do life over?

II. Social

1. What is it like being home for lunch?
2. How do you spend increased time with your wife?
Problems?
3. How did your marriage last a quarter century?
4. What have you learned from your children?
5. Grandchildren?
6. Oldest friend?
7. Activities with others?

III. Psychological

1. Mood over last 6 months?
2. Worries last year?
3. Changes in religious beliefs?

IV. Health

1. Health since retirement?
2. Most annoying aspect of aging?
3. What physical activities have you given up?
4. Cold—what do you do?
5. Days of sick leave?
6. Hospital days total since 1970?
7. Medications?

APPENDIX B

THE INNER CITY MEN AND THE TERMAN WOMEN

In the late 1960s, another important longitudinal study became available to the Harvard Study of Adult Development as a foil for the homogeneous sample of privileged and intelligent white men that made up the College cohort. This study allowed us to compare the College men with a group of white men who were far less privileged, and (at least on IQ testing) less intelligent. A second study became available in the 1980s and offered a comparison with intellectually gifted, but not particularly privileged, women. These opportunities allowed us to draw conclusions about biology versus environment in some of our outcomes, and also, more specifically, about certain kinds of sociological influences. I refer to these two studies throughout the text in contexts where they shed light on, or in some cases expand, the Grant Study findings.

The Inner City Cohort (The Glueck Study of Juvenile Delinquency)

The second cohort joined us in 1969, through the generosity of Sheldon and Eleanor Glueck. Sheldon Glueck was a professor at Harvard Law School and his wife, Eleanor, a groundbreaking social worker there; both were world-famous criminologists. In the 1940s they began an intensive prospective study of 500 white male teenagers from inner-city Boston who had been remanded to reformatories—the Glueck Study of Juvenile Delinquency. They carefully matched them for IQ (their average Wechsler-Bellevue score, based on careful individual testing, was 95) and ethnicity to a control group of 500 boys who had no history of serious delinquency, but who came from the

same high-crime neighborhoods, the same minority identifications, and the same impoverished urban classrooms as the youths who had ended up in reform schools. The Gluecks included no African Americans and no women, and they excluded any boys who by age fourteen had manifested any serious delinquency.

The initial phases of data collection for the Glueck Study were similar to the Grant Study's, with psychiatric interviews, anthropometric measurements, medical examinations, family histories, socioeconomic assessments, and even complete Rorschach tests. But the Gluecks stopped their follow-up in 1962, when their subjects were 32. Thanks to a grant from the National Institute on Alcohol Abuse and Alcoholism, I was able to integrate the two studies in 1970. The control cohort of Inner City men, renamed the *Glueck Study*, is now a part of the Harvard Study of Adult Development, enriching it by the contrast it provides with the socially privileged and high-IQ Harvard men. Like the Grant Study men, the Inner City men deserve great credit for their loyalty to the Study.

Since the integration, the College and Inner City cohorts have been followed in an identical manner, except for the years between 1962 and 1974, when the Inner City men were not followed.

Most of the Inner City men, who were born between 1925 and 1932, had early memories of discrimination and deprivation. But they too were beneficiaries of the G.I. Bill and of America's postwar economic boom. By the 1950s, they, children of once-devalued Irish and Italian immigrant parents, had become the voting majority and political masters of Boston. Their previous distinction as the most disparaged of city minorities had been passed on to recent African-American arrivals from the South. While only one in ten of their fathers belonged to the middle class, at age forty-seven half of the Inner City men had achieved middle-class status. In this achievement the Inner City men were a subsample of underclass "no hope" youth who be-

came solidly middle class—certainly another reasonable definition of “success.”

The Terman Women Cohort

In 1920, Lewis Terman, a renowned Stanford psychologist, started a study of roughly 1,500 grammar-school children. They were mostly 4th graders of a 1906–1911 birth cohort, and they represented all the children in Oakland, San Francisco, and Los Angeles with (carefully tested) IQs of over 140. They have been followed by questionnaires every five years since then, with little attrition, except by death, to this day. Of the Terman group, 672 are women, most of whom went to college.

In 1987, through the generosity of Stanford professors Robert Sears and Albert Hastorf, Caroline Vaillant and I spent a full year reviewing the records of the the 78- to 79-year-old women, and interviewing a representative surviving sample of 40. We used a similar interview to the one we used with the Grant Study men to facilitate comparison. This sample of women who were the Grant Study men’s intellectual peers (or superiors) is the cohort that appears in this volume as the Terman women, and it allowed us to study some of the sociological effects of gender.

In the 1920’s, California was still a young state. One Study member could recall watching the last sailing ships glide through the Golden Gate before the bridge was built. The population of Los Angeles was 500,000. The Terman women were descendants of pioneers. One woman’s grandmother had saved her own life by killing an intruder with a tomahawk. One woman’s father, a high school teacher, used routinely to disarm his students before class. Another father won a stagecoach line to Arizona in a poker game.

Twenty percent of the Terman women’s fathers were in blue-

collar occupations; thirty percent were in “the professions.” Only one Terman woman’s father was an unskilled laborer; he worked as a janitor at the University of California at Berkeley, so that his five bright children could all go to college for free.

The Terman women were physically and intellectually precocious as children. But their high intelligence—they had a mean IQ of 151—did not handicap them psychologically. On the contrary, their mental health was demonstrably better than their classmates’.

After following them until they were 75, Terman and his co-workers observed that the women showed significantly more humor, common sense, perseverance, leadership, and even popularity than their classmates. They were as likely as their classmates to marry, but their physical health was better. At age eighty, the mortality of the Terman women, like the College men’s, has been only half the expected rate for white American women in their birth cohort. As in the Harvard sample, more than half of the Terman women have survived past eighty.

The career situation for these highly intelligent women was full of paradox, however. They grew up with mothers who did not yet have the right to vote. College tuition in California at the time was cheap enough (\$25 to \$50 a term at both Stanford and Berkeley) that a college degree was a realistic expectation for a bright woman. And the Depression, which began when they were twenty, and World War II, which began when they were thirty, put pressures on these women to enter the work force. But the jobs on offer were limited in scope, compensation, and opportunity. When asked what occupational opportunities World War II had opened for her, one Berkeley-educated woman replied dryly, “I finally learned to type.”

Almost half of the Terman women held full-time jobs for most of their lives. Most had gone to college and many to graduate school. Nevertheless, their mean maximum annual income (\$30,000 in 1989

dollars) was identical to that of the Inner City men, who had a mean IQ of 95 and an average of ten to eleven years of education. Wartime demands for Rosie the Riveter and her like might have been an economic boon for high school dropouts, but they were an economic millstone for the gifted Terman women.

For the Terman *men*, it is worth noting, the war really did create great opportunities. The G.I. Bill paid for their graduate schooling and allowed some of them to create the Los Alamos and Livermore laboratories and, ultimately, Silicon Valley. Some of the Terman men gave their brilliance to the Los Angeles entertainment industry. Lewis Terman's own gifted children, a boy and a girl, were both included in his study. Both graduated from Stanford and worked for the university for much of their lives. The son served as provost, and was mentor to many of the founders of Silicon Valley; the daughter was a secretary in one of the dormitories.

Thus, of our three samples, it was the college-educated middle-class Terman women, most of whose relatives had been in the United States for generations, who most clearly illustrated the negative effect of social bigotry upon development.

APPENDIX C

ASSESSMENT OF CHILDHOOD SCALES

1. Child Temperament Scale (Age 0–10)

- 1 = very shy, tics, phobias, bedwetting beyond age 8, dissocial, severe feeding problems, other noted problems.
- 3 = average.
- 5 = good-natured, normally social, an “easy child.”

2. Childhood Environmental Strengths Scale (Age 5–18)

1. Global Impression (rater’s overall hunch)

- 1 = a negative, non-nurturing environment.
- 3 = neither negative nor positive feeling about subject’s childhood.
- 5 = a positive, intact childhood; good relationships with parents, siblings, and others; environment seems conducive to developing self-esteem. A childhood that rater would have wanted.

2. Relationship with Siblings

- 1 = severe rivalry, destructive relationship, sibling undermines child’s self-esteem or no siblings.
- 3 = no good information, not mentioned as good though not particularly bad.
- 5 = close to at least one sibling.

3. Home Atmosphere

- 1 = any noncongenial home, lack of family cohesiveness, parents not together, early maternal separation, known to many social agencies, many moves, financial hardship that impinged greatly on family life.
- 3 = average home: doesn't stand out as good or bad; or lack of information.
- 5 = warm, cohesive atmosphere, parents together, doing things as a family, sharing atmosphere, maternal and paternal presence, few moves, financial stability or special harmony in spite of difficulties.

4. Mother/Child Relationship

- 1 = distant, hostile, blaming others (such as father, teachers) for wrong methods of upbringing, overly punitive, overprotective, expecting too much, mother absent, seductive, not encouraging feeling of self-worth in child.
- 3 = mostly for lack of information or lack of distinct impression about mother.
- 5 = nurturing, encouraging of autonomy, helping boy develop self-esteem, warmth.

5. Father/Child Relationship

- 1 = distant, hostile, overly punitive, expectations unrealistic or not what son wants for himself, paternal absence, negative or destructive relationship.
- 3 = lack of information, no distinct impression about father.
- 5 = warmth, encouraging of autonomy in child, helping to develop self-esteem, do things with son, discusses problems, interested in child.

APPENDIX D

ADULT ADJUSTMENT SCALES

I. Scale for Objective Mental Health from Age 30–50

- | | |
|---|---|
| 1. Income over \$20,000 in 1967 dollars | 1 = Above \$20,000
2 = Below |
| 2. Steady promotion, 1967 | Examination of questionnaires at 5-year intervals from 1946–1967 reveals steady promotion or career progress
1 = Yes
2 = No steady promotion |
| 3. Games, 1967 | Examination of 1951–1967 questionnaires and review of other data reveals games with non-family members (golf, bridge, tennis, etc.)
1 = Yes
2 = No games with others |
| 4. Vacation, 1967 | Evidence from the 1957, 1964, and 1967 questionnaires that the subject took more than two weeks' vacation a year and had fun rather than just dutiful visits to relatives
1 = Yes, takes them
2 = No, ignores vacation |
| 5. Enjoyment of job | Evidence from 1946, 1951, 1954, 1960, 1964, and 1967 questionnaires that the subject enjoyed his job and was enthusiastic about it.
1 = Unambiguous enjoyment
2 = Enjoyment not clear
3 = Definite lack of job enjoyment |
| 6. Psychiatric visits, 1967 | Visits from college through 1967
1 = Under 10 visits
2 = 10 or more visits |

-
- | | |
|----------------------------|--|
| 7. Drug/alcohol use, 1967 | <p>Evidence either that (a) the subject used sleeping pills weekly for a year or tranquilizers and amphetamines daily for a month, or (b) for a period at least as long as a year (or for two points in time) the individual drank more than 8 ounces of alcohol a day or felt he had trouble with control or he, his family, and his friends thought he drank too much</p> <p>1 = No to (a) and (b)
2 = Yes to (a) and/or (b)</p> |
| 8. Days' sick leave 1967 | <p>Based on questionnaires in 1944, 1946, and 1967</p> <p>1 = Less than 5 days
2 = 5 or more days</p> |
| 9. Marital enjoyment, 1967 | <p>Averaging the husband's reports of his marriage in 1954 and 1967 with the wife's report of their marriage in 1967</p> <p>1 = Good (marriage score of 4 or 5)
2 = Intermediate (marriage score of 6 or 7)
3 = Getting or considering a divorce (marriage score of 8+)</p> |
-

II. Scale for Objective Mental Health from Age 50-65

- | | |
|---|--|
| 1. Career
(3 questionnaires) | <p>1 = Working full-time
2 = Significant reduction of work load
3 = Retired</p> |
| 2. Career success
(3 questionnaires) | <p>1 = Current (or pre-retirement) responsibilities/success as great or greater than 1970
2 = Demotions or reduced effectiveness (prior to retirement)</p> |
| 3. Career or retirement enjoyment
(2 questionnaires) | <p>1 = Meaningful, enjoyable
2 = Ambiguous
3 = Working only because he must or feels retirement demeaning/boring</p> |

4. Vacations (2 questionnaires)	1 = 3+ weeks and fun 2 = Less than 3 weeks if working or unplayful retirement
5. Psychiatrist use (2 questionnaires)	1 = No visits 2 = 1–10 visits 3 = Psych hospitalization or 10+ visits
6. Tranquilizer use (2 questionnaires)	1 = None 2 = One use to a month 3 = More than once a month's use
7. Days' sick leave (exclude irreversible illness) (2 questionnaires)	1 = Less than 5 days/year 2 = 5+ days
8. Marriage 1970–1984 (3 questionnaires)	1 = Clearly happy 2 = So-so 3 = Clearly unhappy or divorced
9. Games with others (3 questionnaires)	1 = Regular social activities/sports 2 = Little or none
Total (low score is good)	9–14 = Score compatible with being classified mentally healthy 15–23 = Bottom quartile; excludes individual from being classified as mentally healthy

III. Scale for Objective Mental Health from Age 65–80

- | | |
|-----------------------------------|--|
| 1. Career or retirement enjoyment | 1 = Still enjoying part-time work and/or retirement
2 = Ambiguous or midrange
3 = Dissatisfied with retirement |
| 2. Retirement success, age 65–80 | 1 = Still enjoying part-time work and/or retirement
2 = Ambiguous or midrange
3 = Dissatisfied with retirement |

- | | |
|--|--|
| 3. Contact with younger relatives, age 65–80 | 1 = Meaningful, enjoyable family interaction
2 = Ambiguous or infrequent interaction with young relatives
3 = Avoids or shunned by kids, grandkids, nieces, nephews |
| 4. Use of leisure time, age 65–80 | 1 = Varied, imaginative, and enjoyment of leisure time
2 = Some leisure activities with moderate enjoyment
3 = Bored, passive, unsatisfactory use of leisure time |
| 5. Games with others, age 65–80 | 1 = Many regular social activities: bridge, lunches, golf
2 = Some social activities, but limited involvement.
3 = Almost no social activities |
| 6. Psychiatric use, age 65–80 | 1 = No visits for counseling
2 = 1–10 visits
3 = Psychiatric hospitalization or 10+ visits |
| 7. Mood-altering drug use, age 65–80 | 1 = None
2 = 1–30 days use
3 = More than 1 month's use in a year |
| 8. Marriage, age 65–80 | 1 = Clearly happy (until widowed)
2 = Never married, or so-so or fair if while married
3 = Clearly unhappily married, or divorced with no new intimate relationship |
| 9. Rater's subjective impression, age 65–80 | Rater's subjective impression after reviewing 6–7 questionnaires and other interview data in file
1 = Adjustment to aging is excellent
2 = Adjustment to aging is good or above average
3 = Ambiguous or average adjustment to aging
4 = Poor adjustment to aging
5 = Adjustment to aging worse than for most men |

APPENDIX E

DOMINANT COLLEGE PERSONALITY TRAITS

Dominant College Personality Traits (N=251)

Trait*: (frequency), Definition	Important Correlates
Vital affect: (20%). <i>Expressive, forceful, spontaneous energy, animated</i>	
Sociable, friendly: (22%). <i>Naturally friendly, socially at ease, makes friends easily</i>	
Well integrated: (60%). <i>Steady, stable, dependable, trustworthy, surmounts problems that confront him</i>	Mature defenses—very significant Longevity—very significant Decathlon—very significant Eriksonian maturity—very significant Childhood strengths—significant No depression—significant Stable marriage—significant
Practical, organizing: (37%). <i>Practical not theoretical, organized not analytical, likes getting things done</i>	Conservative—very significant Decathlon—very significant Maturity of defense—very significant Eriksonian maturity—very significant No depression—very significant Stable marriage—significant
Humanistic: (16%). <i>Interested in people, wish to work with people</i>	
Pragmatic: (38%). <i>Practical, conforming, accept the mores of the times</i>	Conservative—very significant No depression—very significant Maturity of defenses—significant
Political: (17%). <i>Interested in government, social reform, public policy rather than people</i>	
Over-integrated, just so: (13%). <i>Neat, meticulous, rigid, depend on routine, systematic</i>	
Bland affect: (38%). <i>Not warm or positive mood, not rich or vital affect</i>	

Trait*: (frequency), Definition	Important Correlates
Self-driving: (14%). <i>Self-control, willpower, persevering, uneasy with leisure</i>	Liberal—very significant
Cultural: (22%). <i>Headed for artistic and literary or at least cultural careers</i>	
Verbalistic: (18%). <i>Facile, lucid, well-formulated and rich in their use of language</i>	
Inarticulate: (14%). <i>Inability to express themselves</i>	
Shy: (18%). <i>Embarrassed, reserved, awkward socially but like people</i>	Liberal—very significant
Physical science: (12%). <i>Mechanical, inductive, like lab work and things more than people</i>	
Sensitive affect: (17%). <i>Shy, subtle, aesthetic, poor adjustment to everyday realities</i>	
Creative and intuitive: (6%). <i>Original, literary, and artistic, spurn concrete forms of thought</i>	
Mood swings: (14%). <i>Strongly marked and/or fluctuations in moods</i>	Liberal—very significant
Inhibited: (19%). <i>Overly moral, indecisive on acting on desires</i>	
Ideational: (21%). <i>Theoretical, analytical, dislike routine, scholarly, prefer literature over science</i>	
Self-conscious, introspective: (25%). <i>More concerned with subjective feelings than others</i>	
Lack of purpose and values: (20%). <i>Drifting, unenthusiastic</i>	Liberal—very significant
Unstable autonomic functions: (14%). <i>Undue anxiety, tremulousness, blushing, sweating, palpitations, functional urinary or GI symptoms</i>	
Asocial: (10%). <i>Other people unimportant, prefer things and their own company</i>	

Trait*: (frequency), Definition	Important Correlates
Incompletely integrated: (15%). <i>Erratic, unreliable, undependable, little perseverance, poorly organized</i>	Decathlon—very significant
Psychopathic: (7%). <i>Confined to a small number of men who might be mentally ill</i>	

*The traits are arranged in the degree that they correlated with the Study’s ABC adjustment. The “soundest” boys most commonly manifested Vital affect and Sociability.

APPENDIX F

STUDY BIBLIOGRAPHY

A. Books

1. Hooton EA: *Young Man, You Are Normal*. New York, G. P. Putnam's Sons, 1945.
2. Heath CW, et al.: *What People Are*. Cambridge, MA, Harvard University Press, 1945.
3. Monks John P: *College Men at War*. Boston, American Academy of Arts and Sciences, 1957.
4. Vaillant GE: *Adaptation to Life*. Boston, MA, Little, Brown, 1977 [reprinted with a new preface in 1995 by Harvard University Press, Cambridge, MA].
5. Vaillant GE: *Natural History of Alcoholism*. Cambridge, MA, Harvard University Press, 1983.
6. Vaillant GE: *Ego Mechanisms of Defense: A Guide for Clinicians and Researchers*. Washington, DC, American Psychiatric Press, 1992.
7. Vaillant GE: *The Wisdom of the Ego*. Cambridge, MA, Harvard University Press, 1993.
8. Vaillant GE: *Natural History of Alcoholism, Revisited*. Cambridge, MA, Harvard University Press, 1995.
9. Vaillant GE: *Aging Well*. Boston, Little, Brown, 2002 [also in Hebrew translation].
10. Vaillant GE: *Spiritual Evolution: A Scientific Defense of Faith*. New York, Doubleday Broadway, 2008.

11. Vaillant GE: *Triumphs of Experience: The Men of the Harvard Grant Study*. Cambridge, MA, Harvard University Press, 2012.

B. Papers

Clark Heath, Director

1. Johnson REL Brouha: Pulse rate, blood lactate and duration of effort in relation to ability to perform strenuous exercise. *Revue Canadienne de Biologie*, 1942, 1, 2, 171–178.
2. Davis, Pauline: Effect on the electroencephalogram of changing the blood sugar level. *Archives of Neurology and Psychiatry*, 1943, 49, 186–194.
3. Wells FL: A research focused upon the normal personality: A note. *Character and Personality*, 1944, 122, 299–301.
4. Wells FL: Mental factors in adjustment to higher education. *Journal of Consulting Psychology*, 1945, 9, 2, 67–86.
5. Savage, Beatrice M: Undergraduate ratings of courses in Harvard College. *Harvard Educational Review*, 1945, 15, 3, 168–172.
6. Seltzer CC: The relationship between the masculine component and personality. *American Journal of Physical Anthropology New Series*, 1945, 3, 33–47.
7. Bock AV: Selection of pre-medical students. *Bios*, 1945, 16, 199–209.
8. Seltzer CC: Chest circumference changes as a result of severe physical training. *American Journal of Physical Anthropology New Series*, 1946, 4, 3, 389–394.
9. Seltzer CC: Body disproportions and dominant personality traits. *Psychosomatic Medicine*, 1946, 8, 2.
10. Heath Clark W, and Lewis W Gregory: Problems of normal

- college students and their families. *School and Society*, 1946, 63, 1638, 355–358.
11. Heath, Clark W, and Lewise W Gregory: What it takes to be an officer. *Infantry Journal*, March 1946.
 12. Wells FL and WL Woods: Outstanding traits: In a selected college group, with some reference to career interests and war records. *Genetic Psychology Monographs*, 1946, 33, 127–249.
 13. Wells FL: Verbal facility: Positive and negative associations. *The Journal of Psychology*, 1947, 23, 3–14.
 14. Wells FL: Verbal excess over quantitation: Two case studies. *The Journal of Psychology*, 1947, 14, 4.
 15. Wells FL: Personal history, handwriting and specific behavior. *The Journal of Psychology*, 1946, 23, 65–82.
 16. Wells FL: Verbal facility, sensitive affect, psychometrics: Two “Palimpsest Personalities.” *The Journal of Psychology*, 1947, 23, 179–191.
 17. Wells FL: Inarticulate, without physical science motivations: Case studies V and VI. *The Journal of Psychology*, 1947, 24, 149–159.
 18. Wells FL: Physical science motivations, inarticulate: Case studies VII and X. *The Journal of Psychology*, 1947, 24, 211–227.
 19. Heath CW, Monks JP, and WL Woods: The nature of career selection in a group of Harvard undergraduates. *Harvard Education Review*, 1947, 17, 3, 190–197.
 20. Heath CW: The hemoglobin of healthy college undergraduates and comparisons with various medical, social, physiologic and other factors. *Blood: The Journal of Hematology*, 1948.
 21. Seltzer CC, FL Wells, and EB McTernan: A relationship be-

- tween Sheldonian somatotype and psychotype. *Journal of Personality*, 1948, 16: 431–436.
22. Wells FL: Verbal aptitudes versus attitudes: Case Studies XI–XIV. *Journal of Genetic Psychology*, 1948, 72, 185–200.
 23. Seltzer CC: Phenotype patterns of racial reference and outstanding personality traits. *Journal of Genetic Psychology*, 1948, 72, 221–245.
 24. Wells FL: Quantitative–spatial aptitudes and motivations: Case Studies XV–XVIII. *Journal of Genetic Psychology*, 1948, 73, 119–140.
 25. Wells FL: Clinical psychology in retrospect and prospect. *Journal of Psychology*, 1949, 27, 125–142.
 26. Wells FL: Adjustment problems at upper extremes of test “intelligence”: Cases XIX–XXVIII. *Journal of Genetic Psychology*, 1949, 74, 61–84.
 27. Wells FL: Psychometric patterns in adjustment problems at upper extremes of test “intelligence”: Cases XXIX–LVI Department of Hygiene Referrals. *Journal of Genetic Psychology*, 1950, 76, 3–37.
 28. Wells FL: Some projective functions of simple geometrical figures: Cases LXXXV–XCV. *Journal of Genetic Psychology*, 1950, 77, 267–281.
 29. Wells FL: Harvard National Scholars in a student frame of reference. *Journal of Genetic Psychology*, 1951, 79, 205–219.
 30. Wells FL: Further notes on Rorschach and case history in Harvard National Scholars: Cases CXIII–CXXVI. *Journal of Genetic Psychology*, 1951, 79, 261–287.
 31. Wells FL: Rorschach and Bernreuter procedures with Harvard

National Scholars in the Grant Study: Cases III, IX, X, XXVII, XXVIII, CII–CXII. *Journal of Genetic Psychology*, 1951, 79, 221–260.

32. Wells FL: The effects of need achievement on the content of TAT stories: A re-examination. *Journal of Abnormal and Social Psychology*, 1953, 48, 532–536.

Charles McArthur, Director

1. McArthur C: Long-term validity of the Strong Interest Test in two subcultures. *Journal of Applied Psychology*, 1954, 38, 346–353.
2. McArthur C: Personalities of public and private school boys. *Harvard Educational Review*, 1954, 24, 256–262.
3. McArthur C: Analyzing the clinical process. *Journal Counseling Psychology*, 1954, 1, 203–206.
4. McArthur C and LB Stevens: The validation of expressed interests as compared with inventoried interests: A fourteen-year follow-up. *Journal of Applied Psychology*, 1955, 39, 184–189.
5. McArthur C: Predictive power of pattern analysis and of job scale analysis of the Strong. *Journal of Counseling Psychology*, 1955, 2, 205–206.
6. McArthur C: Personalities of first and second children. *Psychiatry*, 1956, 19, 47–54.
7. McArthur C: Upper class intelligence as the critical case for a theory of “middle-class bias.” *Journal of Counseling Psychology*, 1957, 4, 23–30.
8. McArthur C: The psychology of smoking (with others). *Journal*

- of *Abnormal and Social Psychology*, March 1958, 56, no. 2, 267–275.
9. McArthur C: Sub-culture and personality during the college years. *Journal of Educational Sociology*, 1960, 33, 260–268.
 10. Dill, David B: Fitness of Harvard men at twenty-two years. *Duke University Council on Gerontology, Proceedings 1961–65*, December 10, 1963, 226–239.
 11. McArthur C: The validity of the Yale Strong Scales at Harvard. *Journal of Counseling Psychology*, 1965, 12, 35–38.
 12. McArthur C: Career choice: It starts at home. *Think*, March–April 1966, 15–18.
 13. Robinson S, Dill DB, Tsankoff SP, Wagner JA, and RD Robinson: Longitudinal studies of aging in 37 men. *Journal of Applied Physiology*, 1975, 38, 263–267.

George Vaillant, Director

1. Vaillant GE, Brighton JR, MacArthur CC: Physicians' use of mood-altering drugs: A 20 year follow-up. *New England Journal of Medicine*, 1970, 282, 365–370.
2. Vaillant GE: Theoretical hierarchy of adaptive ego mechanisms: A 30-year follow-up of 30 men selected for psychological health. (Helene and Felix Deutsch Prize, 1969) *Archives of General Psychiatry*, 1971, 24, 107–118.
3. Vaillant GE: The evolution of adaptive and defensive behaviors during the adult life cycle. Abstracted in *Journal of the American Psychoanalytic Association*, 1971, 19, 110–115.
4. Vaillant GE, MacArthur CC: A thirty-year follow-up of somatic symptoms under emotional stress. In *Life History Research*

- in *Psychopathology*, vol 2. Edited by Roff M, Robins LN, Pollock M. Minneapolis, University of Minnesota Press, 1972.
5. Vaillant GE, Sobawale NC, MacArthur C: Some psychological vulnerabilities of physicians. *New England Journal of Medicine*, 1972, 287, 372-375.
 6. Vaillant GE: Why men seek psychotherapy: I. Results of a survey of college graduates. *American Journal of Psychiatry*, 1972, 129, 645-651.
 7. Vaillant GE, McArthur CC: The natural history of male psychological health: I. The adult life cycle from 18-50. *Seminars in Psychiatry*, 1972; 4:415-427.
 8. Vaillant GE: Antecedents of healthy adult male adjustment. In *Life History Research in Psychopathology*, vol 3. Edited by Roff M, Ricks D. Minneapolis, University of Minnesota Press, 1973.
 9. Vaillant GE, Vaillant CO: The relation of psychological adaptation to medical disability in middle aged men. In *Successful Aging: A Conference Report*. Edited by Pfeiffer E. Durham, NC, Center for the Study of Aging and Human Development, Duke University, 1974.
 10. Vaillant GE: The natural history of male psychological health: II. Some antecedents of healthy adult adjustment. *Archives of General Psychiatry*, 1974, 31, 15-22.
 11. Robinson S, Dill DB, Tsankoff SP, Wagner JA, and RD Robinson: Longitudinal studies of aging in 37 men. *Journal of Applied Physiology*, 1975, 38, 263-267.
 12. Vaillant GE: The natural history of male psychological health: III. Empirical dimensions of mental health. *Archives of General Psychiatry*, 1975, 32, 420-426.
 13. Vaillant GE: Natural history of male psychological health: V.

- The relation of choice of ego mechanisms of defense to adult adjustment. *Archives of General Psychiatry*, 1976, 33, 535–545.
14. Vaillant GE: Natural history of male psychological health: IV. What kinds of men do not get psychosomatic illness. *Psychosomatic Medicine*, 1978, 40, 420–431.
 15. Vaillant GE: Natural history of male psychological health: VI. Correlates of successful marriage and fatherhood. *American Journal of Psychiatry*, 1978, 135, 653–659.
 16. Vaillant GE: Natural history of alcoholism: I. A preliminary report. In *Human Functioning in Longitudinal Perspective*. Edited by Sells SB, Crandall R, Roff M, Strauss JS, Pollin W. Baltimore, MD, Williams and Wilkins, 1979.
 17. Vaillant GE: Health consequences of adaptation to life. *American Journal of Medicine*, 1979, 67, 732–734.
 18. Vaillant GE: Natural history of male psychological health: Effects of mental health on physical health. *New England Journal of Medicine*, 1979, 301, 1249–1254.
 19. Vaillant GE, Milofsky ES: Natural history of male psychological health: IX. Empirical evidence for Erikson's model of the life cycle. *American Journal of Psychiatry*, 1980, 137, 1348–1359.
 20. Vaillant GE: Adolf Meyer was right: Dynamic psychiatry needs the life chart. Strecker Monograph Series, #16. *The Journal of the National Association of Private Psychiatric Hospitals*, 1980, 11, 4–14.
 21. Vaillant GE: Natural history of male psychological health: VIII. Antecedents of alcoholism and “orality.” *American Journal of Psychiatry*, 1980, 137, 181–186.

22. Vaillant GE: The doctor's dilemma. In *Alcoholism Treatment in Transition*. Edited by Grant M. London, Croom Helm, 1980.
23. Vaillant GE: Paths out of alcoholism. In *Evaluation of the Alcoholic: Implications for Research, Theory, and Treatment*. Edited by Meyer RE et al. Maryland, National Institute on Alcohol Abuse and Alcoholism, 1981.
24. Vaillant GE, Vaillant CO: Natural history of male psychological health: X. Work as a predictor of positive mental health. *American Journal of Psychiatry*, 1981, 138, 1433-1440.
25. Vaillant GE, Gale L, Milofsky ES: Natural history of male alcoholism: II. The relationship between different diagnostic dimensions. *Journal of Studies on Alcohol*, 1982, 43, 216-232.
26. McAdams DP, Vaillant GE: Intimacy, motivation and psychosocial adjustment. *Journal of Personality Assessment*, 1982, 46, 586-593.
27. Vaillant GE, Milofsky ES: The etiology of alcoholism: A prospective viewpoint. *American Psychologist*, 1982, 37, 494-503.
28. Vaillant GE, Milofsky E: The natural history of male alcoholism: IV. Paths to recovery. *Archives of General Psychiatry*, 1982, 39, 127-133.
29. Vaillant GE: Childhood environment and maturity of defense mechanisms. In *Human Development: An Interactional Perspective*. Edited by Magnusson D, Allen V. New York, Academic Press, 1983.
30. Vaillant GE, Clark W, Cyrus C, Milofsky ES, Kopp J, Wulsin V, Mogielnicki NP: Prospective study of alcoholism treatment: Eight-year follow-up. *American Journal of Medicine*, 1983, 75, 455-463.

31. Vaillant GE: Natural history of male alcoholism: V. Is alcoholism the cart or the horse to sociopathy? *British Journal of Addiction*, 1983, 78, 317–326.
32. Long JVF, Vaillant GE: Natural history of male psychological health: XI. Escape from the underclass. *American Journal of Psychiatry*, 1984, 141, 341–346.
33. Hartmann E, Milofsky E, Vaillant G, Oldfield M, Falke R, Ducey C: Vulnerability to schizophrenia: Prediction of adult schizophrenia using childhood information. *Archives of General Psychiatry*, 1984, 41, 1050–1056.
34. Beardslee WR, Vaillant GE: Prospective prediction of alcoholism and psychopathology. *Journal of Studies on Alcohol*, 1984, 45, 500–503.
35. Vaillant GE: The Study of Adult Development at Harvard Medical School. In *Handbook of Longitudinal Research*, vol. 2. Edited by Mednick SA, Harway M, Finello KM. New York, Praeger, 1984.
36. Vaillant GE: An empirically derived hierarchy of adaptive mechanisms and its usefulness as a potential diagnostic axis. *Acta Psychiatrica Scandinavica*, 1985; Supplementum no. 319, 71, 171–180.
37. Drake RE, Vaillant GE: A validity study of Axis II of DSM-III. *American Journal of Psychiatry*, 1985, 142, 553–558.
38. Snarey J, Vaillant GE: How lower- and working-class youth become middle-class adults: The association between ego defense mechanisms and upward social mobility. *Child Development*, 1985, 56, 899–910.
39. Vaillant GE, Bond M, Vaillant CO: An empirically validated hi-

- erarchy of defense mechanisms. *Archives of General Psychiatry*, 1986, 43, 786–794.
40. Vaillant GE: Cultural factors in the etiology of alcoholism: A prospective study, in *Alcohol and Culture: Comparative Perspectives from Europe and America*, Annals of the New York Academy of Sciences, vol. 472. Edited by Babor TF. New York, The New York Academy of Sciences, 1986.
 41. McCullough L, Vaillant C, Vaillant GE: Toward reliability in identifying ego defenses through verbal behavior. In *Empirical Studies of Ego Mechanisms of Defense*. Edited by Vaillant GE. Washington, DC, American Psychiatric Press, 1986.
 42. Vaillant GE, Vaillant CO: A cross-validation of two empirical studies of defenses. In *Empirical Studies of Ego Mechanisms of Defense*. Edited by Vaillant GE. Washington, DC, American Psychiatric Press, 1986.
 43. Beardslee WR, Son L, Vaillant GE: Exposure to parental alcoholism during childhood and outcome in adulthood: A prospective longitudinal study. *British Journal of Psychiatry*, 1986; 149:584–591.
 44. Vaillant GE: Time: An important dimension of psychiatric epidemiology. In *Psychiatric Epidemiology: Progress and Prospects*. Edited by Cooper B. London, Croom Helm, 1987.
 45. Felsman JK, Vaillant GE: Resilient children as adults: A forty year study. In *The Invulnerable Child*. Edited by Anthony EJ, Cohler BJ. New York, Guilford Press, 1987.
 46. Vaillant GE: An empirically derived hierarchy of adaptive mechanisms and its usefulness as a potential diagnostic axis. In *Diagnosis and Classification in Psychiatry: A Critical Appraisal of*

- DSM-III*. Edited by Tischler GL. New York, Cambridge University Press, 1987.
47. Phillips KA, Vaillant GE, Schnurr P: Some physiologic antecedents of adult mental health. *American Journal of Psychiatry*, 1987; 144:1009–1013.
 48. Vaillant C, Milofsky E, Richards R, Vaillant GE: A social case-work contribution to understanding alcoholism. *Health and Social Work*, 1987; 12:169–176.
 49. Vaillant GE: A developmental view of old and new perspectives of personality disorders. *Journal of Personality Disorders*, 1987; 1:146–156.
 50. Vaillant GE, McCullough L: The Washington University Sentence Completion Test compared with other measures of adult ego development. *American Journal of Psychiatry*, 1987; 144:1189–1194.
 51. Snarey J, Son L, Kuehne VS, Hauser S, Vaillant G: The role of parenting in men's psychosocial development: A longitudinal study of early adulthood infertility and midlife generativity. *Developmental Psychology*, 1987; 23:593–603.
 52. Vaillant GE: Some differential effects of genes and environment on alcoholism. In *Alcoholism: Origins and Outcome*. Edited by Rose RM, Barrett JE. New York, Raven Press, 1988.
 53. Drake RE, Vaillant GE: Predicting alcoholism and personality disorder in a 33-year longitudinal study of children of alcoholics. *British Journal of Addiction*, 1988; 83:799–807.
 54. Drake RE, Vaillant GE: Introduction: Longitudinal views of personality disorder. *Journal of Personality Disorders*, 1988; 2:44–48.

55. Drake RE, Adler DA, Vaillant GE: Antecedents of personality disorders in a community sample of men. *Journal of Personality Disorders*, 1988; 2:60–68.
56. Vaillant GE, Schnurr P: What is a case? A 45-year study of psychiatric impairment within a college sample selected for mental health. *Archives of General Psychiatry*, 1988; 45:313–319.
57. Peterson C, Seligman MEP, Vaillant GE: Pessimistic explanatory style is a risk factor for physical illness: A thirty-five year longitudinal study. *Journal of Personality and Social Psychology*, 1988; 55:23–27.
58. Vaillant GE: What can long-term follow-up teach us about relapse and prevention of relapse in addiction. *British Journal of Addiction*, 1988; 83:1147–1157.
59. Vaillant GE: Attachment, loss and rediscovery. *Hillside Journal of Clinical Psychiatry*, 1988; 10:148–164.
60. Vaillant GE: The evolution of defense mechanisms during the middle years. In *The Middle Years: New Psychoanalytic Perspectives*. Edited by Oldham JM, Liebert RS. New Haven, Yale University Press, 1989.
61. Vaillant GE, Vaillant CO: Natural history of male psychological health: XII. A 45-year study of successful aging at age 65. *American Journal of Psychiatry*, 1990; 147:31–37.
62. Schnurr PP, Vaillant CO, Vaillant GE: Predicting exercise in late midlife from young adult personality characteristics. *International Journal of Aging and Human Development*, 1990; 30:153–161.
63. Vaillant GE: Repression in college men followed for half a century. In *Repression and Dissociation: Implications for Personality*

- Theory, Psychopathology, and Health*. Edited by Singer JL. Chicago, University of Chicago Press, 1990.
64. Vaillant GE: Avoiding negative life outcomes: Evidence from a forty-five year study. In *Successful Aging: Perspectives from the Behavioral Sciences*. Edited by Baltes PB, Baltes MM. Cambridge, Cambridge University Press, 1990.
 65. Vaillant GE, Vaillant CO: Determinants and consequences of creativity in a cohort of gifted women. *Psychology of Women Quarterly*, 1990; 14:607–616.
 66. Vaillant GE: Prospective evidence for the effects of environment upon alcoholism. In *Alcoholism and the Family: The 4th International Symposium of the Psychiatric Research Institute of Tokyo*. Edited by Saitoh S, Steinglass P, Schuckit MA. New York, Brunner-Mazel, 1992.
 67. Vaillant GE, Schnurr PP, Baron JA, Gerber PD: A prospective study of the effects of cigarette smoking and alcohol abuse on mortality. *Journal of General Internal Medicine*, 1991; 6:299–304.
 68. Vaillant GE: The association of ancestral longevity with successful aging. *Journal of Gerontology*, 1991; 46:292–298.
 69. Vaillant GE, Milofsky ES: The etiology of alcoholism: A prospective viewpoint. In *Society, Culture, and Drinking Patterns Re-examined*. Edited by Pittman, DJ, White, HR. New Brunswick, NJ, Rutgers Center of Alcohol Studies, 1991.
 70. Vaillant GE: Physician, cherish thyself: The hazards of self-prescribing. *Journal of the American Medical Association*, 1992; 267:2373–2374.
 71. Vaillant GE, Roston D, McHugo GJ: An intriguing association

- between ancestral mortality and male affective disorder. *Archives of General Psychiatry*, 1992; 49:709–715.
72. Vaillant CO, Vaillant GE: Is the U-curve of marital satisfaction an illusion? A 40-year study of marriage. *Journal of Marriage and the Family*, 1993; 55:230–239.
73. Vaillant GE, Koury SH: Late midlife development. In *The Course of Life*. Edited by Pollock GH, Greenspan SI. Madison, CT, International Universities Press, 1993.
74. Vaillant GE: Is alcoholism more often the cause or the result of depression? *Harvard Review of Psychiatry*, 1993; 1:94–99.
75. Vaillant GE: Successful aging and psychosocial well-being. In *Older Men's Lives*. Edited by Thompson EH. Thousand Oaks, California, Sage Publications, 1994.
76. Vaillant GE: Ego mechanisms of defense and personality psychopathology. *Journal of Abnormal Psychology*, 1994; 103:44–50.
77. Vaillant GE: La créativité chez les hommes et les femmes “ordinaires.” *Nervuure*, 1994; 7:25–27.
78. Lee KA, Vaillant GE, Torrey WC, Elder GH: A 50-year prospective study of the psychological sequelae of World War II combat. *American Journal of Psychiatry*, 1995; 152:516–522.
79. Vaillant GE: Addictions over the life course. In *Psychotherapy, Psychological Treatments and the Addictions*. Edited by Edwards G, Dare C. Cambridge, England, Cambridge University Press, 1996.
80. Vaillant GE: A long-term follow-up of male alcohol abuse. *Archives of General Psychiatry*, 1996; 53:243–249.
81. Vaillant GE, Orav J, Meyer SE, McCullough-Vaillant L, Ros-

- ton D: Late life consequences of affective spectrum disorder. *International Psychogeriatrics*, 1996; 8:1–20.
82. Cui X, Vaillant GE: The antecedents and consequences of negative life events in adulthood: A longitudinal study. *American Journal of Psychiatry*, 1996; 152:21–26.
 83. Vaillant GE, Gerber PD: Natural history of male psychological health: XIII. Who develops high blood pressure and who responds to treatment. *American Journal of Psychiatry*, 1996; 153:24–29.
 84. Cui X, Vaillant GE: Does depression generate negative life events? *Journal of Nervous and Mental Disease*, 1997; 185:145–150.
 85. Vaillant GE: The natural history of alcoholism and its relationship to liver transplantation. *Liver Transplantation and Surgery*, 1997; 3:304–310.
 86. Vaillant GE: Poverty and paternalism: A psychiatric viewpoint. In *The New Paternalism: Supervisory Approaches to Poverty*, 279–304. Edited by L. Meade. Washington, D.C., Brookings, 1998.
 87. Vaillant GE: Natural history of male psychological health: XIV. Relationship of mood disorder vulnerability to physical health. *American Journal of Psychiatry*, 1998; 155:184–191.
 88. Soldz S, Vaillant GE: A 50-year longitudinal study of defense use among inner city men: A validation of the DSM-IV defense axis. *Journal of Nervous and Mental Disease*, 1998; 186:104–111.
 89. Vaillant GE, Meyer SE, Mukamal K, Soldz S: Are social supports in late midlife a cause or a result of successful physical aging? *Psychological Medicine*, 1998; 28:1159–1168.
 90. Wulsin LR, Vaillant GE, Wells VE: A systematic review of the mortality of depression. *Psychosomatic Medicine*, 1999; 61:6–17.

91. Vaillant GE: Lessons learned from living. *Scientific American Presents*, 1999; 10:32–37.
92. Soldz S, Vaillant GE: The big five personality traits and the life course: A 45-year longitudinal study. *Journal of Research in Personality*, 1999; 33:208–232 [1999 Best Article].
93. Laub JH, Vaillant GE: Delinquency and mortality: A 50-year follow-up study of 1000 delinquent and nondelinquent boys. *American Journal of Psychiatry*, 2000; 157:96–102.
94. Vaillant GE: Adaptive mental mechanisms: Their role in a positive psychology. *American Psychologist*, 2000; 55:89–98.
95. Vaillant GE: Prevention of alcoholism: Reflections of a naturalist. In *Childhood Onset of Adult Psychopathology*. Ed. by JL Rapoport. Washington, DC, American Psychiatric Press, 2000.
96. Vaillant GE, Davis JT: Social/emotional intelligence and midlife resilience in schoolboys with low tested intelligence. *American Journal of Orthopsychiatry*, 2000; 70:215–222.
97. Vaillant GE: If addiction is involuntary, how can punishment help? In *Drug Addiction and Drug Policy*. Edited by PB Heymann and WN Brownsberger. Cambridge, MA, Harvard University Press, 2001, pp. 144–167.
98. Cui X and Vaillant GE: Stressful life events and late adulthood adaptation. In *Aging in Good Health*. Edited by SE Levkoff, YK Cheej, and S Noguchi. New York, Springer, 2001.
99. Vaillant GE and K Mukamal: Successful aging. *American Journal of Psychiatry*, 2001; 158:839–847.
100. Vaillant GE: The value of a hierarchy of defenses. In *Entwicklung und Risiko*. Edited by G. Roper, G. Noam, and C. von Hagen. Berlin, Kohlhammer, 2001.

101. Vaillant GE, Vaillant CO: The Study of Adult Development. In *Landmark Studies in the Twentieth Century*. Edited by Anne Colby and Erin Phelps. New York, Russell Sage Foundation, 2002.
102. Vaillant GE: Healthy aging among inner-city men. *International Psychogeriatrics*, 2002; 13:425–437.
103. Vaillant GE: Natural history of addiction and pathways to recovery. In *Principles of Addiction Medicine*, 3rd edition. Edited by AW Graham et al., Chevy Chase, MD, American Society of Addiction Medicine, 2003.
104. Vaillant GE: Mental health. *American Journal of Psychiatry*, 2003; 160:1373–1384.
105. Vaillant GE: A 60-year follow-up of male alcoholism. *Addiction*, 2003; 98:1043–1051.
106. Isaacowitz DM, Vaillant GE, Seligman MEP: Strength and satisfaction across the adult lifespan. *International Journal of Aging and Human Development*, 2003; 27:181–201.
107. Vaillant GE: Mental health. In *Comprehensive Textbook of Psychiatry*, 8th edition. Edited by BJ Sadock and VA Sadock. Philadelphia, Williams and Wilkins, 2004.
108. Vaillant GE: Positive aging. In *Positive Psychology in Practice*. Edited by Linley PA and Joseph S. Hoboken, NJ: John Wiley & Sons, 2004.

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1. Vaillant GE, Batalden MB, Orav J, Roston D, Barrett JE: Evidence for an X-linked personality trait related to affective illness. *Australian and New Zealand Journal of Psychiatry*, 2005; 39:730–735.

2. Vaillant GE: Alcoholics Anonymous: Cult or cure. *Australian and New Zealand Journal of Psychiatry*, 2005; 39: 431–436.
3. Vaillant GE, DiRago AC, Mukamal K: Natural history of male psychological health XV: Retirement satisfaction. *American Journal of Psychiatry*, 2006; 163:682–688.
4. Vaillant GE, DiRago AC: Satisfaction with retirement in men's lives. In *Charting a New Life: The Crown of Life Dynamics of the Early Retirement Period*. Edited by J James and P Wink. New York: Springer, 2006.
5. Waldinger RJ, Vaillant GE, Orav EJ: Childhood sibling relationships as predictors of major depression in adulthood: A thirty-year prospective study. *American Journal of Psychiatry*, 2006; 164:949–954.
6. Vaillant GE: Generativity—A form of unconditional love. In *Altruism and Health*. Edited by S. Post. New York, Oxford University Press, 2007.
7. DiRago AC, Vaillant GE (2007). Resilience in Inner City youth: Childhood predictors of occupational status across the lifespan. *Journal of Youth and Adolescence*, 36:61–70.
8. Vaillant GE, Templeton J, Ardel M, Meyer S: Natural history of male mental health: Health and religious involvement. *Social Science and Medicine*, 2008, 66:221–231.
9. Koenig L and Vaillant GE: A prospective study of church attendance and health over the lifespan. *Health Psychology*, 28, 2009:117–124.
10. Frosch ZA, Dierker LC, Rose JS, and Waldinger RJ: Smoking trajectories, health, and mortality across the adult lifespan. *Addictive Behaviors*, 2009, 34:701–704.
11. Beardslee WR, Vaillant GE: Adult development. In *Psychiatry*,

- 2nd ed. Edited by Tasman A, Kay J, Lieberman J. New York, Saunders, 2008.
12. DiRago A, Vaillant GE: Mature defense mechanisms. In *The Encyclopedia of Positive Psychology*. Edited by SJ Lopez. 2:600–605. Malden, MA, Wiley-Blackwell, 2009.
 13. Grenfield EA, Vaillant GE, Marks NF: Do formal religious participation and spiritual perceptions have independent linkages with diverse dimensions of psychological well-being? *Journal of Health and Social Behavior*, 2009; 50:196–212.
 14. Vaillant GE, Vaillant CO: Normality and mental health. In *Comprehensive Textbook of Psychiatry*, 9th edition. Edited by BJ Sadock, VA Sadock, and P. Ruiz. Philadelphia, Williams and Wilkins, 2009.
 15. Martin-Joy J and George E. Vaillant: Recognizing and promoting resilience. In *Successful Cognitive and Emotional Aging*. Edited by C Depp and DV Jeste. Arlington VA: American Psychiatric Press, 2010.
 16. Katie A McLaughlin, Laura D Kubzansky, Erin C Dunn, Robert Waldinger, George Vaillant, and Karestan C Koenen: Childhood social environment, emotional reactivity to stress, and mood and anxiety disorders across the life course. *Depression and Anxiety*, 2010; 27: 1087–1094 (2010).
 17. Waldinger RA and MS Schulz: Facing the music or burying our heads in the sand? Adaptive emotion regulation in midlife and late life. *Research in Human Development*, 2010; 7:292–306.
 18. Waldinger RJ, Schulz MS: What's love got to do with it? Social connections, perceived health stressors, and daily mood in married octogenarians. *Psychology and Aging*, 2010, 25: 422–431.
 19. Pergakis MB, Hasan NS, Heller NR, Waldinger RJ: Octoge-

- narian reports of lifetime spiritual experiences: Types of experience and early life predictors. *Journal of Religion, Spirituality, and Aging*, 2010; 22: 220–238.
20. Waldinger RJ, Schulz MS: Linking hearts and minds in couple interactions: Intentions, attributions and overriding sentiments. *Journal of Family Psychology*, 2006, 20:494–504.
 21. Waldinger RJ, Kensinger EA, Schulz MS: Neural activity, neural connectivity, and the processing of emotionally-valenced information in older adults: Links with life satisfaction. *Cognitive, Affective & Behavioral Neuroscience*, 2011, 11:426–436.
 22. Liu L, Cohen S, Schulz MS, Waldinger RJ: Sources of somatization: Exploring the roles of insecurity in relationships and styles of anger experience and expression. *Social Science & Medicine*, 2011, 73:1436–1443.
 23. Cohen S, Schulz MS, Weiss E, Waldinger RJ: “A” for effort: The individual and dyadic contributions of empathic accuracy and empathic effort to relationship satisfaction. *Journal of Family Psychology*, 2012, 26:236–245.